Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

ZH

Selected Domain

Domain 2: Orbyte Smart SME Challenge

Selected SDG Track (1–17)

SDG 8: Decent Work and Economic Growth

Project / Solution Name

ChaseFlow for Orbyte (working name)

Name (Leader)

Wong Zong Han

Email & Phone (Leader)

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University Affiliation(s)

Asia Pacific University of Technology & Innovation (APU)

Number of Team Members

1 (solo)

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro, small and medium enterprises (MSMEs) are the backbone of the economy: the Department of Statistics Malaysia (DOSM) reports that they added 39.5% of GDP and employed 48.7% of all workers in 2024 [1]. Yet many act as the bank for their customers. Experian's State of Credit report shows SMEs in Malaysia took an average of 65 days to pay in January to May 2024, against 57 days for large corporations [2]. That measures how long payment takes, not how late it is.

Every unpaid invoice is money already spent on stock, wages or rent, and the owner who must chase it also does everything else. Chasing means opening a spreadsheet in the evening, trying to remember who was reminded last time, and writing WhatsApp messages one by one in the customer's language. Three things go wrong: reminders come late or are skipped; the tone is wrong, too soft for a habitual late payer or too harsh for a loyal customer who forgot once; and promises such as "I will pay on Friday" live in memory or chat history, so a broken promise is noticed late.

Cash-flow stress is what small firms can least absorb, so better collections support SDG 8: growing small enterprises and the jobs they carry. The routine described is the team's working assumption, not yet validated by interviews with SME owners.

Sources: [1] DOSM, MSMEs Performance 2024 (31 July 2025). [2] Experian Malaysia, State of Credit, May 2024.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ChaseFlow is a working prototype that helps an SME owner decide whom to chase, what to say and in which language. It follows one journey: trigger, connected data, decision, human approval, visible outcome.

1. "Check Overdue" lists every invoice that is past its due date and still unpaid.
2. Each invoice is joined with its customer (language, phone), payment history and any payment promise.
3. Plain-code rules set a risk tier (low, medium, high) that the owner can check by hand. Google Gemini then explains the risk in plain English, recommends an action and drafts the reminder in English, Bahasa Malaysia or Chinese. It may move the tier by one level and must say why; the screen shows the rule tier and the AI tier separately.
4. The owner edits and approves. Sending is simulated, so nothing leaves the app.
5. The approval is written to an Activity Log. When a customer promises a date, the owner records it; the app works out from the real payment whether it was kept, paid late or broken, and the next draft takes that into account.

Where AI adds real value: reading a customer's history in context and writing the message in the right tone and language. Where it does not: the risk rule, the promise status and all amounts are ordinary code. If Gemini fails, a rule-based template keeps the app usable. Workshop skills used: calling a GenAI API (Gemini), from the SDC Phase 1 GenAI workshop. Hosting is on Replit. The prototype has no database yet; it uses in-memory synthetic data. All data is made up and Orbyte is not connected. Out of scope: real payments, sending real messages, legal or financial advice.

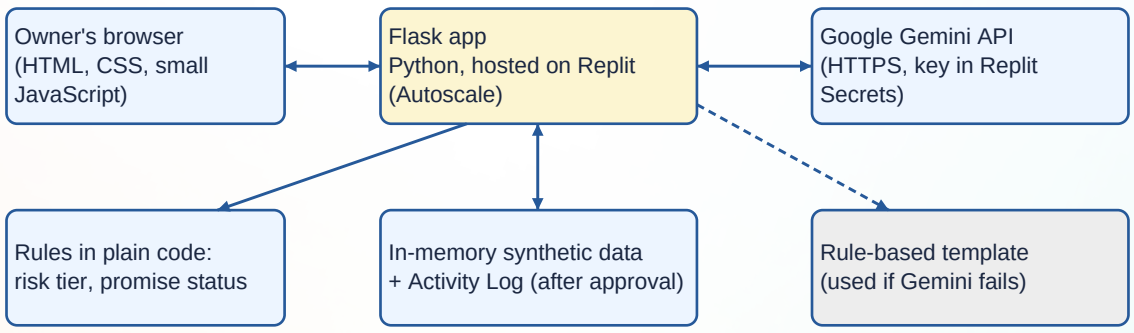
3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies: Python 3 and Flask (backend, routes, Jinja2 templates); HTML, CSS and small vanilla JavaScript (interface, charts drawn as inline SVG); Google Gemini API, model gemini-flash-lite-latest, called with the requests library; in-memory synthetic data (no database yet); Replit workspace and Autoscale deployment; GitHub for source code. Live prototype: https://chaseflow-orbyte.replit.app



System block diagram. Sending is simulated: nothing leaves the app.

Sponsor-provided resources: Replit Core (provided by the organisers through the SDC portal, and used) for the development workspace and for hosting the live prototype. The RunPod GPU credit is not used, because the only AI component is the Gemini API, which runs on Google's servers.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define. The team leader studies Accounting, so receivables and collections are familiar ground, and likes finding SME problems to solve. The user is the owner-manager who chases late invoices by hand. The working assumption is that the job is "decide whom to chase and what to say"; it is not yet tested in interviews with SME owners.
Ideate. Ideas were brainstormed together with AI assistants and narrowed to one complete journey. Recovery forecasts, auto-reading WhatsApp replies and partial payments were left out.
Prototype and test. Built in small steps and tried in the running app and on the live site. The core rules have 13 automated tests; AI failures fall back to a template.
Iterate. Payment-promise tracking was added so reminders reflect what the customer already promised.
Hardest risk and fallback. AI output that is wrong or unavailable. The rule sets the baseline, the AI must explain any change, a human approves every message, and a template takes over if Gemini fails.
Deployment. Already live on Replit. Before real use it needs a database, invoice import from Orbyte or another accounting tool, and a PDPA and consent review before any real message is sent.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG 8, Target 8.3:** "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services." Secondary: Target 8.2, higher economic productivity through technological upgrading and innovation. Source: United Nations, Sustainable Development Goal 8 targets, sdgs.un.org/goals/goal8.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**65 days:** the average time Malaysian SMEs took to make a payment in January to May 2024, compared with 57 days for large corporations (Experian Malaysia, State of Credit, May 2024). This is a published benchmark for SMEs in general, not a measurement of our users, and it measures time to pay, not lateness. **Our own starting point (assumption):** about 5 minutes for the owner to check the history and write one reminder by hand in one language. We will check this by timing the team leader on 5 invoices before the final presentation.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	**Minutes from "invoice found overdue" to "reminder approved", per invoice.** Measured by timing the demo flow on the 15 demo invoices; the Activity Log records when each reminder is approved. **Long-term outcome (not tested in Phase 1):** average days to pay for invoices that were chased, compared with the 65-day SME benchmark. This needs a pilot with real SMEs.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**At least 80% less time:** from the assumed 5 minutes by hand to 1 minute or less per reminder, measured on the 15 demo invoices by the final prototype presentation. **Longer-term ambition, not a Phase 2 claim:** move SME time to pay from 65 days toward the 57 days of large corporations (about 12% shorter), which can only be shown with real users.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable: no earlier code, design or research is included.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Flask (with Jinja2, Werkzeug and other dependencies)	Code: Python web framework and libraries	BSD-3-Clause (blinker: MIT)
requests	Code: HTTPS calls to the Gemini API	Apache-2.0
Inter typeface (Google Fonts)	Code: font served by Google Fonts	SIL Open Font License 1.1
Gemini Flash-Lite (Google)	Model: pre-trained, called through the Gemini API, not modified	Google Gemini API Terms of Service
No other third-party code, dataset or model	The demo data is synthetic and written by the team	None

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

WONG

[Date]

23/10/26